

Ch1 Manifolds (continued)

Second Countability & Partition of Unity

LXJ

September 22, 2026

Paracompactness

Covers, refinements and local finiteness

Definition (Cover, subcover, refinement)

A collection $\{U_\alpha\}$ is a *cover* of $W \subset M$ if $W \subset \bigcup_\alpha U_\alpha$. It is an *open cover* if each U_α is open. A subcollection of $\{U_\alpha\}$ which still covers is called a *subcover*. A *refinement* $\{V_\beta\}$ of the cover $\{U_\alpha\}$ is a cover such that

$$\forall \beta, \exists \alpha \text{ s.t. } V_\beta \subset U_\alpha.$$

Definition (Locally finite)

A collection $\{A_\alpha\}$ of subsets of M is *locally finite* if whenever $m \in M$, $\exists$ a neighborhood W_m of m s.t.

$$W_m \cap A_\alpha \neq \emptyset \text{ for only finitely many } \alpha.$$

Paracompactness

Definition (Paracompact)

A topological space is *paracompact* if every open cover has a *locally finite refinement*.

Proposition

Let X be a topological space which is

- *locally compact* (each point has a compact neighborhood),
- *Hausdorff*, and
- *second countable*.

Then X is *paracompact*.

Corollary

Every differentiable manifold is paracompact.

Proof sketch of the proposition

Sketch.

1. **Exhaustion.** X has a countable basis $\{B_n\}_{n \in \mathbb{N}^+}$ of open sets with compact closure. Inductively build $G_1 = B_1$, $G_{k+1} = \bigcup_{n=1}^{j_{k+1}} B_n$ with j_{k+1} smallest s.t. $\overline{G_k} \subset \bigcup_{n=1}^{j_{k+1}} B_n$. Then

$$G_1 \subset G_2 \subset \cdots \subset G_i \subset G_{i+1} \subset \cdots, \quad \bigcup_k G_k = X.$$

2. **Refinement.** Given an open cover $\{U_\alpha\}$: $\overline{G_i} \setminus G_{i-1}$ is compact, covered by finitely many $U_\alpha \cap (G_{i+1} \setminus \overline{G_{i-2}})$; similarly cover $\overline{G_2}$ by finitely many $U_\alpha \cap G_3$. Collecting these gives a countable refinement $\{V_n\}$.
3. **Local finiteness.** Each $p \in X$ lies in at most two “rings” $\overline{G_i} \setminus G_{i-1}$, each meeting only finitely many V_n .



An exercise hidden in the proof

Exercise

If X is second countable, locally compact and Hausdorff, then X has a countable basis consisting of open sets with compact closure.

Remark

The picture behind the construction:

$$\overline{G_i} \setminus G_{i-1} \subset G_{i+1} \setminus \overline{G_{i-2}} \quad (\text{compact ring inside open ring}).$$

Each V_n has compact closure, being contained in $\overline{G_{i+1}} \setminus G_{i-2}$ or $\overline{G_3}$.

Partition of Unity

Partition of unity

Definition (Partition of unity)

A *partition of unity* on M is a collection $\{\varphi_i \mid i \in I\}$ of C^∞ functions on M s.t.

(a) the collection $\{\text{supp } \varphi_i\}_{i \in I}$ is *locally finite*;

(b) $\sum_{i \in I} \varphi_i = 1$ on M , and $\varphi_i(p) \geq 0$ for all $p \in M$ and $i \in I$.

Definition (Subordination)

A partition of unity $\{\varphi_i \mid i \in I\}$ is *subordinated to the cover* $\{U_\alpha\}_{\alpha \in A}$ if

$$\forall i \in I, \exists \alpha \in A \quad \text{s.t.} \quad \text{supp } \varphi_i \subset U_\alpha.$$

It is subordinated to $\{U_i\}_{i \in I}$ with the same index if $\text{supp } \varphi_i \subset U_i$ for each $i \in I$.

Bump functions on $\mathbb{R}^d$

Proposition (Existence of bump functions)

There exists a non-negative C^∞ function φ on $\mathbb{R}^d$ which equals 1 on the closed cube $\overline{C(1)}$ and zero on the complement of the open cube $C(2)$.

Similarly, $\exists$ a non-negative C^∞ function φ on $\mathbb{R}^d$ which equals 1 on the closed ball $\overline{B_0(1)}$ and zero outside of the open ball $B_0(2)$.

Sketch.

1. $f(t) = \begin{cases} e^{-1/t^2}, & t > 0 \\ 0, & t \leq 0 \end{cases}$ is C^∞ , and $f(t) \rightarrow 1^-$ as $t \rightarrow +\infty$.
2. $g(t) = \frac{f(t)}{f(t) + f(1-t)}$ is C^∞ (denominator never zero), with $g(t) \equiv 0$ ($t \leq 0$) and $g(t) \equiv 1$ ($t \geq 1$).
3. $h(t) = g(t+2)g(2-t)$: then $h \equiv 1$ on $[-1, 1]$ and $h \equiv 0$ outside $[-2, 2]$.
4. General case: $\varphi = (h \circ r_1) \cdots (h \circ r_d)$ for the cube, or $\varphi = h(\|x\|^2)$

Existence of Partition of Unity

Existence of partition of unity

Theorem (Existence of partition of unity)

Let M be a differentiable manifold and $\{U_\alpha\}_{\alpha \in A}$ an open cover of M .

1. There exists a countable partition of unity $\{\varphi_i \mid i = 1, 2, \dots\}$ subordinate to the cover $\{U_\alpha\}$, with $\text{supp } \varphi_i$ compact for each i .
2. If one does not require compact support, there exists a partition of unity $\{\varphi_\alpha\}_{\alpha \in A}$ with the same index (subordinate to $\{U_\alpha\}$, $\text{supp } \varphi_\alpha \subset U_\alpha$), with at most countably many of the φ_α not identically zero.

Proof sketch: local bumps

Sketch.

1. Let $\{G_i\}$ exhaust M as in the previous lemma, set $G_0 = \emptyset$. For $\forall p \in M$, let i_p be the largest integer s.t. $p \in M \setminus \overline{G_{i_p}}$. Choose α_p s.t. $p \in U_{\alpha_p}$.
2. Take a coordinate system (V, τ) centered at p with

$$V \subset U_{\alpha_p} \cap (G_{i_p+2} \setminus \overline{G_{i_p}}), \quad \tau(V) \supset \overline{C(2)}.$$

3. Define

$$\psi_p = \begin{cases} \varphi \circ \tau, & \text{on } V, \\ 0, & \text{elsewhere,} \end{cases}$$

where φ is the bump function of the previous lemma. Then $\psi_p \in C^\infty(M)$ and $\psi_p \equiv 1$ on a neighborhood W_p of p .



Proof sketch: normalization and regrouping

Sketch (continued).

4. For each $i \geq 1$, finitely many W_p cover the compact ring $\overline{G_i} \setminus G_{i-1}$. Ordering all these ψ_p in a sequence ψ_j : $\{\text{supp } \psi_j\}$ is locally finite, so

$$\psi = \sum_{j=1}^{\infty} \psi_j$$

is a well-defined C^∞ function on M with $\psi(p) > 0$ for all $p \in M$.

5. Then $\varphi_j = \psi_j/\psi$ forms the desired partition of unity with $\text{supp } \varphi_j$ compact.
6. For the same-index version, set $\tilde{\varphi}_\alpha = 0$ if no $\text{supp } \varphi_j \subset U_\alpha$, otherwise the sum of all such φ_j . Local finiteness gives

$$\text{supp } \tilde{\varphi}_\alpha \subset U_\alpha.$$

□

Corollary: separating a closed set from an open set

Corollary

Let $G \subset M$ be open and $A \subset M$ closed with $A \subset G$. Then $\exists$ a C^∞ function $\varphi : M \rightarrow [0, 1]$ s.t.

- (a) $0 \leq \varphi(p) \leq 1$ for all $p \in M$;
- (b) $\varphi \equiv 1$ on A ;
- (c) $\text{supp } \varphi \subset G$.

Proof.

Take a partition of unity $\{\phi, \psi\}$ subordinate to the cover $\{G, M \setminus A\}$ with $\text{supp } \phi \subset G$ and $\text{supp } \psi \subset M \setminus A$. Then $\varphi = 1 - \psi$ is the desired function. □